



Seep Pahuja
Biology Educator
(NEET UG)



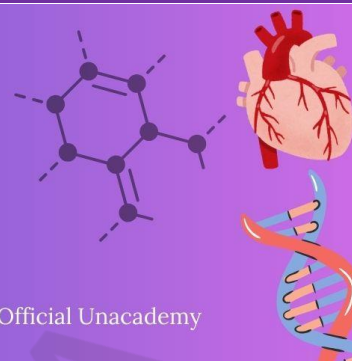
seep.pahuja



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LAST 10 YEARS

NEET PYQ



Chapter 1 : Sexual Reproduction in Flowering Plants

1. Given below are two statements :

[NEET-2025]

Assertion (A) : A typical unfertilised, angiosperm embryo sac at maturity is 8 nucleate and 7-celled.

Reason (R) : The egg apparatus has 2 polar nuclei. In the light of the above statements, choose the correct answer from the options given below :

- (1) Both A and R are true and R is the correct explanation of A
- (2) Both A and R are true but R is NOT the correct explanation of A
- (3) A is true but R is false
- (4) A is false but R is true?

2. Given below are two statements :

[NEET-2025]

Assertion (A) : Both wind and water pollinated flowers are not very colourful and do not produce nectar. Reason (R) : The flowers produce enormous amount of pollen grains in wind and water pollinated flowers. In the light of the above statements, choose the correct answer from the options given below:

- (1) Both A and R are true and R is the correct explanation of A
- (2) Both A and R are true but R is NOT the correct explanation of A
- (3) A is true but R is false
- (4) A is false but R is true

3. How many meiotic and mitotic divisions need to occur for the development of a mature female gametophyte from the megaspore mother cell in an angiosperm plant?

[NEET-2025]

- | | |
|-----------------------------|------------------------------|
| (1) 2 Meiosis and 3 Mitosis | (2) 1 Meiosis and 2 Mitosis |
| (3) 1 Meiosis and 3 Mitosis | (4) No Meiosis and 2 Mitosis |

4. Given below are two statements : [NEET-2025]

Assertion (A) : Cells of the tapetum possess dense cytoplasm and generally have more than one nucleus. Reason (R) : Presence of more than one nucleus in the tapetum increases the efficiency of nourishing the developing microspore mother cells.

- (1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is NOT the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

5. Match List I with List II : [NEET-2025]

List I

- A. Scutellum
B. Non-albuminous seed
C. Epiblast
D. Perisperm

List II

- I. Persistent nucellus
II. Cotyledon of Monocot seed
III. Groundnut
IV. Rudimentary cotyledon

Choose the option with all correct matches.

- (1) A-II, B- III, C-IV, D-I
(2) A-IV, B- III, C-II, D-I
(3) A-IV, B- III, C-I, D-II
(4) A-II, B- IV, C-III, D-I

6. Which part of the ovule stores reserve food materials ? [Re-NEET-2024]

- (1) Nucellus
(2) Integument
(3) Placenta
(4) Funicle

7. Pollen grains remain preserved as fossils due to the presence of : [Re-NEET-2024]

- (1) Epidermal layer
(2) Tapetum
(3) Exine layer
(4) Intine layer

8. Identify the set of correct statements: [NEET-2024]

- A. The flowers of Vallisneria are colourful and produce nectar.
B. The flowers of water lily are not pollinated by water.
C. In most of water-pollinated species, the pollen grains are protected from wetting.
D. Pollen grains of some hydrophytes are long and ribbon like.
E. In some hydrophytes, the pollen grains are carried passively inside water.

Choose the correct answer from the options given below.

- (1) C, D and E only
(2) A, B, C and D only
(3) A, C, D and E only
(4) B, C, D and E only

9. Identify the correct description about the given figure:

[NEET-2024]



- (1) Wind pollinated plant inflorescence showing flowers with well exposed stamens.
(2) Water pollinated flowers showing stamens with mucilaginous covering.
(3) Cleistogamous flowers showing autogamy.
(4) Compact inflorescence showing complete autogamy
10. In which of the following sets of families, the pollen grains are viable for months?
(1) Rosaceae, Liliaceae and Poaceae [NEET-2023-MANIPUR]
(2) Leguminosae, Solanaceae and Rosaceae
(3) Solanaceae, Poaceae and Liliaceae
(4) Brassicaceae, Liliaceae and Poaceae
11. Transfer of pollen grains from anther to stigma of another flower of same plant is known as:
(1) Autogamy (2) Cleistogamy [NEET-2023-MANIPUR]
(3) Geitonogamy (4) Xenogamy
12. In angiosperms the correct sequence of events in formation of female gametophyte in the ovule is
(A) 3 successive free nuclear divisions in functional megaspores. [NEET-2023-MANIPUR]
(B) Degeneration of 3 megaspores.
(C) Meiotic division in megaspore mother cell.
(D) Migration of 3 nuclei towards each pole.
(E) Formation of wall resulting in seven celled embryo sac.
Choose the correct answer from the options given below.
(1) (B), (C), (A), (D), (E) (2) (C), (B), (A), (D), (E)
(3) (A), (B), (C), (D), (E) (4) (C), (E), (A), (D), (B)
13. What is the function of tassels in the corn cob? [NEET-2023]
(1) To attract insects (2) To trap pollen grains
(3) To disperse pollen grains (4) To protect seeds
14. In angiosperm, the haploid, diploid and triploid structures of a fertilized embryo sac sequentially are : [NEET-2023]
(1) Synergids, Primary endosperm nucleus and zygote
(2) Antipodals, synergids, and primary endosperm nucleus
(3) Synergids, Zygote and Primary endosperm nucleus
(4) Synergids, antipodals and Polar nuclei

15. Large, colourful, fragrant flowers with nectar are seen in [NEET-2023]
(1) Insect pollinated plants (2) Bird pollinated plants
(3) Bat pollinated plants (4) Wind pollinated plants
16. Given below two statements : One is labelled as Assertion A and the other is labelled as Reason R :
Assertion A : A flower is defined as modified shoot wherein the shoot apical meristem changes to floral meristem. [NEET-2023]
Reason R : Internode of the shoot gets condensed to produce different floral appendages laterally at successive node instead of leaves.
In the light of the above statements, choose the correct answer from the options given below :
(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is NOT the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true
17. Identify the incorrect statement related to Pollination:- [NEET-2022]
(1) Pollination by wind is more common amongst abiotic pollination
(2) Flowers produce foul odours to attract flies and beetles to get pollinated
(3) Moths and butterflies are the most dominant pollinating agents among insects
(4) Pollination by water is quite rare in flowering plants
18. Given below are two statements:- [NEET-2022]
Statement I : Cleistogamous flowers are invariably autogamous
Statement II : Cleistogamy is disadvantageous as there is no chance for cross pollination.
(1) Both Statement I and Statement II are incorrect
(2) Statement I is correct but Statement II is incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct
19. The residual persistent part which forms the perisperm in the seeds of beet is :- [Re-NEET-2022]
(1) Calyx (2) Endosperm
(3) Nucellus (4) Integument
20. Which of the following can be expected if scientists succeed in introducing apomictic gene into hybrid varieties of crops ? [Re-NEET-2022]
(1) Polyembryony will be seen and each seed will produce many plantlets
(2) Seeds of hybrid plants will show longer dormancy
(3) Farmers can keep on using the seeds produced by the hybrids to raise new crop year after year
(4) There will be segregation of the desired characters only in the progeny

21. To ensure that only the desired pollens fall on the stigma in artificial hybridization process:-

[Re-NEET-2022]

- (a) the female flower buds of plant producing unisexual flower need not be bagged.
- (b) there is no need to emasculate unisexual flowers of selected female parent
- (c) emasculated flowers are to be bagged immediately after cross pollination
- (d) emasculated flowers are to be bagged after removal of anthers
- (e) bisexual flowers, showing protogyny are never selected for cross

Choose the correct answer from the options given below :

- (1) (a), (b) and (c) only
- (2) (b), (c) and (d) only
- (3) (b), (c) and (e) only
- (4) (a), (d) and (e) only

22. The term used for transfer of pollen grains from anthers of one plant to stigma of a different plant which, during pollination, brings genetically different types of pollen grains to stigma, is :-

- (1) Xenogamy
 - (2) Geitonogamy
 - (3) Chasmogamy
 - (4) Cleistogamy
- [NEET-2021]

23. A typical angiosperm embryo sac at maturity is :-

[NEET-2021]

- (1) 8-nucleate and 7-celled
- (2) 7-nucleate and 8-celled
- (3) 7-nucleate and 7-celled
- (4) 8-nucleate and 8-celled

24. In some members of which of the following pairs of families, pollen grains retain their viability for months after release ?

[NEET-2021]

- (1) Poaceae ; Rosaceae
- (2) Poaceae; Leguminosae
- (3) Poaceae; Solanaceae
- (4) Rosaceae ; Leguminosae

25. In water hyacinth and water lily, pollination takes place by :-

[NEET-2020]

- (1) insects and water
- (2) insects or wind
- (3) water currents only
- (4) wind and water

26. The body of the ovule is fused within the funicle at:-

[NEET-2020]

- (1) Chalaza
- (2) Hilum
- (3) Micropyle
- (4) Nucellus

27. The plant parts which consist of two generations one within the other :-

[NEET-2020]

- (a) Pollen grains inside the anther
- (b) Germinated pollen grain with two male gametes
- (c) Seed inside the fruit
- (d) Embryo sac inside the ovule

- (1) (a) and (d)
- (2) (a) only
- (3) (a), (b) and (c)
- (4) (c) and (d)

28. Which of the following is incorrect for wind pollinated plants ? [NEET-2020(COVID-19)]
- (1) Well exposed stamens and stigma
 - (2) Many ovules in each ovary
 - (3) Flowers are small and not brightly coloured
 - (4) Pollen grains are light and non-sticky
29. Vegetative propagules in Agave is termed as :- [NEET-2020(COVID-19)]
- (1) Rhizome
 - (2) Bulbil
 - (3) Offset
 - (4) Eye
30. Male and female gametophytes do not have an independent free living existence in :- [NEET-2020(COVID-19)]
- (1) Pteridophytes
 - (2) Algae
 - (3) Angiosperms
 - (4) Bryophytes
31. In some plants, the female gamete develops into embryo without fertilization. This phenomenon is known as :- [NEET-2019]
- (1) Autogamy
 - (2) Parthenocarpy
 - (3) Syngamy
 - (4) Parthenogenesis
32. Which one of the following statements regarding post-fertilization development in flowering plants is incorrect ? [NEET-2019]
- (1) Ovary develops into fruit
 - (2) Zygote develops into embryo
 - (3) Central cell develops into endosperm
 - (4) Ovules develop into embryo sac
33. Persistent nucellus in the seed is known as :- [NEET-2019]
- (1) Chalaza
 - (2) Perisperm
 - (3) Hilum
 - (4) Tegmen
34. What is the fate of the male gametes discharged in the synergid? [NEET-2019]
- (1) One fuses with the egg, other(s) degenerate(s) in the synergid.
 - (2) All fuse with the egg.
 - (3) One fuses with the egg, other(s) fuse(s) with synergid nucleus.
 - (4) One fuses with the egg and other fuses with central cell nuclei.
35. Which is the most common type of embryo sac in angiosperms ? [NEET-2019(Odisha)]
- (1) Tetrasporic with one mitotic stage of divisions
 - (2) Monosporic with three sequential mitotic divisions
 - (3) Monosporic with two sequential mitotic divisions
 - (4) Bisporic with two sequential mitotic divisions

36. What type of pollination takes place in Vallisneria? [NEET-2019(Odisha)]

- (1) Pollination occurs in submerged condition by water
- (2) Flowers emerge above surface of water, and pollination occurs by insects.
- (3) Flowers emerge above water surface, and pollen is carried by wind.
- (4) Male flowers are carried by water currents to female flowers at surface of water

37. In which one of the following, both autogamy and geitonogamy are prevented?

- (1) Wheat (2) Papaya [NEET-2019(Odisha)]
- (3) Castor (4) Maize

38. Which of the following flowers only once in its life-time ? [NEET-2018]

- (1) Bamboo species (2) Jackfruit
- (3) Mango (4) Papaya

39. Pollen grains can be stored for several years in liquid nitrogen having a temperature of:-

- (1) -120°C (2) -80°C [NEET-2018]
- (3) -196°C (4) -160°C

40. Double fertilization is :- [NEET-2018]

- (1) Fusion of two male gametes of a pollen tube with two different eggs
- (2) Fusion of one male gamete with two polar nuclei
- (3) Fusion of two male gametes with one egg
- (4) Syngamy and triple fusion

41. Which one of the following plants shows a very close relationship with a species of moth, where none of the two can complete its life cycle without the other? [NEET-2018]

- (1) Hydrilla (2) Yucca
- (3) Banana (4) Viola

42. Functional megaspore in an angiosperm develops into ? [NEET-2017]

- (1) Endosperm (2) Embryo sac
- (3) Embryo (4) Ovule

43. Attractants and rewards are required for :- [NEET-2017]

- (1) Entomophily (2) Hydrophily
- (3) Cleistogamy (4) Anemophily

44. Plants which produce characteristic pneumatophores and show vivipary belong to: [NEET-2017]

- (1) Halophytes (2) Psammophytes
- (3) Hydrophytes (4) Mesophytes

45. Flowers which have single ovule in the ovary and are packed into inflorescence are usually pollinated by:- [NEET-2017]
- (1) Bee (2) Wind
(3) Bat (4) Water
46. A dioecious flowering plant prevents both :- [NEET-2017]
- (1) Autogamy and geitonogamy (2) Geitonogamy and xenogamy
(3) Cleistogamy and xenogamy (4) Autogamy and xenogamy
47. Double fertilization is exhibited by :- [NEET-2017]
- (1) Algae (2) Fungi
(3) Angiosperms (4) Gymnosperms
48. The coconut water from tender coconut represents:- [NEET-I 2016]
- (1) Endocarp (2) Fleshy mesocarp
(3) Free nuclear proembryo (4) Free nuclear endosperm
49. Proximal end of the filament of stamen is attached to the:- [NEET-I 2016]
- (1) Anther (2) Connective
(3) Placenta (4) Thalamus or petal
50. Which one of the following statements is not true? [NEET-I 2016]
- (1) Tapetum helps in the dehiscence of anther
(2) Exine of pollen grains is made up of sporopollenin
(3) Pollen grains of many species cause severe allergies
(4) Stored pollen in liquid nitrogen can be used in the crop breeding programmes
51. Cotyledon of maize grain is called :- [NEET-I 2016]
- (1) plumule (2) coleorhiza
(3) coleoptile (4) scutellum
52. Seed formation without fertilization in flowering plants involves the process of :- [NEET-I 2016]
- (1) Sporulation (2) Budding
(3) Somatic hybridization (4) Apomixis
53. Which of the following statements is not correct? [NEET-I 2016]
- (1) Pollen grains of many species can germinate on the stigma of a flower, but only one pollen tube of the same species grows into the style.
(2) Insects that consume pollen or nectar without bringing about pollination are called pollen/nectar robbers.
(3) Pollen germination and pollen tube growth are regulated by chemical components of pollen interacting with those of the pistil.

(4) Some reptiles have also been reported as pollinators in some plant species.

54. Which one of the following generates new genetic combinations leading to variation ?

- | | | |
|-----------------------------|---------------------------|----------------|
| (1) Sexual reproduction | (2) Nucellar polyembryony | [NEET-II 2016] |
| (3) Vegetative reproduction | (4) Parthenogenesis | |

55. In majority of angiosperms: - [NEET-II 2016]

- (1) Reduction division occurs in the megaspore mother cells
- (2) A small central cell is present in the embryo sac
- (3) Egg has a filiform apparatus
- (4) There are numerous antipodal cells

56. Pollination in water hyacinth and water lily is brought about by the agency of :- [NEET-II 2016]

- | | |
|-----------|---------------------|
| (1) Birds | (2) Bats |
| (3) Water | (4) Insects or wind |

57. The ovule of an angiosperm is technically equivalent to :- [NEET-II 2016]

- | | |
|---------------------------|--------------------|
| (1) Megaspore mother cell | (2) Megaspore |
| (3) Megasporangium | (4) Megasporophyll |

ANSWER KEY

1. (3)
2. (2)
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- 56. (4)
- 57. (3)

SEEPP PAKHUA